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import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")

df = pd.read_csv("/content/Sample - Superstore.csv", encoding='latin1')

print("Dataset Loaded Successfully!")

print("\nFirst 5 Rows:")
print(df.head())

print("\nDataset Shape:")
print(df.shape)

print("\nColumns:")
print(df.columns.tolist())

print("\nMissing Values:")
print(df.isnull().sum())

df["Order Date"] = pd.to_datetime(df["Order Date"])

region_sales = (
    df.groupby("Region")["Sales"]
    .sum()
    .sort_values(ascending=False)
)

plt.figure(figsize=(10, 6))

region_sales.plot(kind="bar")

plt.title("Total Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.xticks(rotation=0)

plt.show()

monthly_sales = (
    df.groupby(df["Order Date"].dt.to_period("M"))["Sales"]
    .sum()
)

monthly_sales.index = monthly_sales.index.astype(str)

plt.figure(figsize=(14, 6))

plt.plot(
    monthly_sales.index,
    monthly_sales.values,
    marker="o"
)

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.xticks(rotation=45)

plt.tight_layout()

plt.show()

plt.figure(figsize=(10, 6))

plt.hist(
    df["Sales"],
    bins=30
)
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plt.title("Distribution of Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")

plt.show()

plt.figure(figsize=(10, 6))

sns.boxplot(
    data=df,
    x="Region",
    y="Sales"
)

plt.title("Sales Distribution by Region")
plt.xlabel("Region")
plt.ylabel("Sales")

plt.show()

plt.figure(figsize=(10, 6))

sns.scatterplot(
    data=df,
    x="Sales",
    y="Profit"
)

plt.title("Sales vs Profit")
plt.xlabel("Sales")
plt.ylabel("Profit")

plt.show()

numeric_df = df.select_dtypes(
    include=["int64", "float64"]
)

correlation = numeric_df.corr()

plt.figure(figsize=(10, 8))

sns.heatmap(
    correlation,
    annot=True,
    fmt=".2f",
    cmap="coolwarm"
)

plt.title("Correlation Heatmap")

plt.show()

category_sales = (
    df.groupby("Category")["Sales"]
    .sum()
    .sort_values(ascending=False)
)

plt.figure(figsize=(10, 6))

sns.barplot(
    x=category_sales.index,
    y=category_sales.values
)

plt.title("Sales by Product Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")

plt.show()

category_profit = (
    df.groupby("Category")["Profit"]
    .sum()
    .sort_values(ascending=False)
)
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plt.figure(figsize=(10, 6))

sns.barplot(
    x=category_profit.index,
    y=category_profit.values
)

plt.title("Profit by Product Category")
plt.xlabel("Category")
plt.ylabel("Total Profit")

plt.show()

segment_sales = (
    df.groupby("Segment")["Sales"]
    .sum()
    .sort_values(ascending=False)
)

plt.figure(figsize=(10, 6))

sns.barplot(
    x=segment_sales.index,
    y=segment_sales.values
)

plt.title("Sales by Customer Segment")
plt.xlabel("Customer Segment")
plt.ylabel("Total Sales")

plt.show()

print("\n=====")
print("VISUALIZATION ANALYSIS COMPLETED")
print("=====")

print("1. Bar chart shows sales comparison across regions.")
print("2. Line chart shows monthly sales trends.")
print("3. Histogram shows the distribution of sales.")
print("4. Box plot identifies sales outliers.")
print("5. Scatter plot shows the relationship between sales and profit.")
print("6. Correlation heatmap shows relationships among numerical variables.")
print("7. Category analysis compares sales and profit across products.")
```

Dataset Loaded Successfully!

First 5 Rows:

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335
4	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335

	Customer Name	Segment	Country	City
0	Claire Gite	Consumer	United States	Henderson
1	Claire Gite	Consumer	United States	Henderson
2	Darrin Van Huff	Corporate	United States	Los Angeles
3	Sean O'Donnell	Consumer	United States	Fort Lauderdale
4	Sean O'Donnell	Consumer	United States	Fort Lauderdale

	Postal Code	Region	Product ID	Category	Sub-Category
0	42420	South	FUR-BO-10001798	Furniture	Bookcases
1	42420	South	FUR-CH-10000454	Furniture	Chairs
2	90036	West	OFF-LA-10000240	Office Supplies	Labels
3	33311	South	FUR-TA-10000577	Furniture	Tables
4	33311	South	OFF-ST-10000760	Office Supplies	Storage

	Product Name	Sales	Quantity
0	Bush Somerset Collection Bookcase	261.9600	2
1	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.9400	3
2	Self-Adhesive Address Labels for Typewriters b...	14.6200	2
3	Bretford CR4500 Series Slim Rectangular Table	957.5775	5
4	Eldon Fold 'N Roll Cart System	22.3680	2

	Discount	Profit
0	0.00	41.9136
1	0.00	219.5820
2	0.00	6.8714
3	0.45	-383.0310
4	0.20	2.5164

[5 rows x 21 columns]

Dataset Shape:

(9994, 21)

Columns:

['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode', 'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'St

Missing Values:

Row ID	0
Order ID	0
Order Date	0
Ship Date	0
Ship Mode	0
Customer ID	0
Customer Name	0
Segment	0
Country	0
City	0
State	0
Postal Code	0
Region	0
Product ID	0
Category	0
Sub-Category	0
Product Name	0
Sales	0
Quantity	0
Discount	0
Profit	0

dtype: int64



